

Alain Dzaul Contreras Olea

SDET Quality Assurance

Contact Information

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- **Location:** Mexico City, Mexico

Key Achievements

Metric	Achievement
125%	Quality improvement (4/10 → 9/10)
800+	Automated tests created
100%	FDA audit pass rate
5	Bug-free releases in a row

Professional Summary

- Specialized in QA for pharmaceutical-grade software under FDA 21CFR Part 11 compliance
- Proven track record: 5 consecutive zero-bug releases, 125% quality improvement (4/10 → 9/10)
- Full-stack QA expertise: 800+ automated tests across React/TypeScript/Node.js/Next.js/Nest.js stack
- Technical leadership: Led QA for 7 developers, implemented CI/CD, Three Amigos, Definition of Ready
- Multi-industry experience: Pharmaceutical, Enterprise Software, Energy sectors

Professional Experience

Quality Assurance | Rubidex

Mexico City | February 2025 - Present

Key Responsibilities:

- Improved product quality from 6/10 to 8/10 client satisfaction (33% increase)
- Built 800+ automated tests (600 unit, 200 integration, 10 E2E) serving dual purpose: validation and living documentation
- Led QA for 7 developers across 2 teams, blocking 5+ critical bugs pre-production

Additional Details:

- Implemented CI/CD pipelines for Frontend/Backend and agile processes: Three Amigos, Design Handoffs, Definition of Ready

- Acted as technical bridge between C-Level and development, refining requirements and defining acceptance criteria for complex modules
 - Established change control governance and process standardization to protect sprint scope and mitigate risks
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Software Quality Engineer | AspenTech

Mexico City | March 2021 - July 2024

Key Responsibilities:

- 5 consecutive releases with zero critical bugs (zero showstoppers)
- Passed 2 FDA audits + 3+ client audits with no critical observations
- Managed 1,600+ test cases for pharmaceutical spectrometry software (50 specialized users)

Additional Details:

- Delivered 7-8 successful releases for enterprise client with 3 interdependent products
 - Complete documentation under FDA 21CFR Part 11 and ISO 9001 compliance
 - Hybrid testing approach: 10% automated (critical flows) + 90% manual (regulatory flexibility)
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Software Engineer | Infosys

Mexico City | July 2019 - February 2021

Key Responsibilities:

- L2/L3 technical support for 5,000+ users in Fortune 500 client (multi-continental operations)
- Resolved ~25 tickets/month in Oracle ERP Supply Chain using Python, SQL, and Java
- Collaborated in multicultural team for international enterprise-scale project

Additional Details:

- Basic Android development with Kotlin for internal tools
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Engineering Intern | Pemex

Mexico City, Tula Hidalgo, Salina Cruz Oaxaca | September 2018 - March 2019

Key Responsibilities:

- Analyzed and collected data from gas production optimization system, performing ethylene production forecasting with predictions that accurately reflected subsequent national production decline
 - Visited 5 facilities (refineries, distribution centers, gas processing plants) for operational audits
 - Supported gas and liquid transportation process tracking
 - Gained exposure to large-scale industrial operations in energy sector
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Education

Escuela Superior de Ingeniería Química e Industrias Extractivas (IPN)

Mexico City Bachelor's degree in Petroleum Chemical Engineering

Instituto Tecnológico de Buenos Aires (ITBA)

Buenos Aires, Argentina Exchange Program

Courses in Chemical/Industrial Engineering:

- Process Simulation and Optimization
 - Quality Management
 - Environmental Engineering
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Technical Skills

AI & Development Tools

- **Antigravity** - AI-powered development environment for rapid prototyping
- **Claude Code** - AI coding assistant CLI for autonomous software development
- **Gemini CLI** - Google's AI assistant command-line interface for coding tasks
- **WindSurf** - AI-first IDE with intelligent code completion and refactoring

Programming Languages

- **TypeScript** - Typed superset of JavaScript for safer, scalable applications
- **JavaScript** - Dynamic programming language for web and server-side development
- **Python** - Versatile language for automation, testing, and data analysis
- **SQL** - Query language for relational database operations and analytics
- **Java** - Object-oriented language for enterprise applications
- **C++** - High-performance language for system and scientific computing

Testing & QA

- **Jest** - JavaScript testing framework for unit and integration tests
- **Playwright** - End-to-end testing framework for web applications across browsers
- **Vitest** - Fast unit test framework optimized for Vite/modern projects
- **SuperTest** - HTTP assertion library for testing Node.js REST APIs
- **Manual Testing** - Exploratory testing with human judgment for UX and edge cases
- **White-box Testing** - Internal structure testing with full code knowledge

Frontend & Backend

- **React** - Component-based UI library for building interactive interfaces
- **Next.js** - React framework with SSR, routing, and full-stack capabilities
- **Nest.js** - Progressive Node.js framework with TypeScript and modular architecture
- **Node.js** - JavaScript runtime for building scalable server-side applications

Code Quality & CI/CD

- **ESLint** - JavaScript linter for identifying code quality and style issues
- **Prettier** - Opinionated code formatter for consistent style across teams
- **GitHub Actions** - Automation platform for CI/CD workflows in GitHub repositories
- **Git** - Distributed version control system for tracking code changes
- **Azure DevOps** - Microsoft's DevOps platform for CI/CD, repos, and project management

Industry Software

- **Aspen HYSYS** - Process simulation software for chemical engineering
- **Ranorex** - GUI test automation framework for desktop and web applications

- **AutoCAD** - Computer-aided design software for 2D/3D technical drawings
 - **Postman** - API development and testing platform for REST/GraphQL
 - **OPC Tools** - Industrial automation protocol tools for SCADA systems
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Certifications & Licenses

- **Certified in ISO 9001:2015**
 - **Professional License** - Number: 12075687
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Languages

- **Spanish** - Native
 - **English** - Professional Fluency
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